Decomposing Prompt Style: A Feature-Level Analysis of LLM Behavior

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Abstract

We conduct a controlled factorial experiment to examine how fine-grained prompt style features - formality, politeness, uncertainty, and length - influence LLM response behavior across five models: GPT, Claude Sonnet, DeepSeek, Gemini, and Mistral. Using a 2×2×2×3 design across 14 base questions, we collected and analyzed 1,680 responses on token length, formality, and politeness. We find that prompt length and formality are the strongest drivers of response length, while politeness and certainty remain largely fixed regardless of prompt framing - consistent with RLHF instilling strong stylistic defaults. Models exhibit asymmetric style mirroring: informal prompts are matched 75.8 percent of the time vs 33 percent for formal ones, and polite prompts are mirrored 64 percent of the time compared to 22-26 percent for neutral or impolite ones. Model identity also proves a stronger predictor of verbosity than prompt style, with DeepSeek averaging twice the tokens of Sonnet. These results suggest that stylistic prompt engineering has real but bounded effects, and that model selection may matter more than prompt tone for communication-sensitive applications.

1 Introduction

LLMs have become integral to both human-AI and AI-AI communication, yet we lack understanding of prompt style, independent of semantic context, stylistically affects LLM responses and behaviors. In our project, we extended the findings from "Do LLMs Behave Differently When the Prompter is Human vs Another LLM?" The original IdeaHub implementation tested between two styles of prompts - human versus LLM, with manual prompt creation based on character defining features from HC3 (Guo et al., 2023); however, modern LLMs no longer definitively follow these characteristics. As LLMs become more integrated into daily communication, humans are increasingly adopting LLM-like language patterns (formal structures, explicit instructions, politeness markers), while LLMs are being fine-tuned to sound more natural and conversational. This convergence makes the binary human/LLM distinction increasingly artificial and less useful for understanding what actually drives differential model behavior. Thus, we tested how fine-grained style features (formality, length, and politeness) influence LLM stylistic behavior on different Q&A prompts across various different LLMs (Mistral, Gemini, GPT, Sonnet, and DeepSeek). As for extensions of the previous work, we extended the previous work by experimenting with five models as the previous work focused on GPT-4.1 mini and Claude Sonnet 4 to see the extent to which certain behaviors generalize across different LLMs. Furthermore, we focus our evaluation metrics on stylistic features like certainty and politeness, which were not tested in the previous work. Obtaining further understanding of these stylistic fine-grained features will allow us to curate goal-specific prompt engineering and design more effective inter-agent communication protocols.

2 Related Work

"The politeness Package; Detecting Politeness in Natural Language" Michael Yeomans et al. (2018)

This article detects politeness in English natural language, defining it as how well a speaker makes the listener feel respected and autonomous. The authors developed 36 politeness markers based on words, phrases, and sentiments to measure a document's politeness — similar measures to those we apply when evaluating prompt politeness.

"Prompt Perturbations Reveal Human-Like Biases in LLM Survey Responses" Jens Rupprechet et al. (2025)

This paper examines whether prompt perturbations (e.g., typos) undermine LLM robustness on closed-ended survey questions. Our project similarly studies perturbations but uses open-ended questions rather than fixed answer sets, adding complexity to the task.

"A Human-AI Comparative Analysis of Prompt Sensitivity in LLM-Based Relevance Judgment" Arabzadeh & Clarke et al. (2025)

This study tests 90 human- and LLM-generated prompts across three relevance tasks using GPT-4o, LLaMA, and Mistral, finding that LLM-generated prompts outperform human ones and that GPT-4o is less sensitive to prompt variation than smaller models. Our work extends this by also examining response structure and format, and by introducing greater prompt variation.

"Concise Thoughts: Impact of Output Length on LLM Reasoning and Cost" Sania Nayab et al. (2024)

This paper introduces Constrained Chain-of-Thought (CCoT), a prompt engineering technique that reduces output length while maintaining accuracy compared to standard Chain-of-Thought (CoT). Like our project, it compares two prompt techniques; unlike ours, it focuses solely on conciseness rather than broader stylistic dimensions such as formality and politeness.

"A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT" Jules White et al. (2023)

White et al. (2023) catalog prompting strategies for ChatGPT, showing that prompt structure - beyond semantic content - meaningfully affects model behavior. However, the work emphasizes explicit instructions over implicit stylistic cues. Our project examines more granular stylistic features across multiple models, offering broader insights into effective prompt engineering.

3 Methods

Overview of Experimental Setup: We conducted a controlled factorial experiment to isolate the casual effects of individual prompt style features on LLM response behavior. Our approached comprised of four phases: (1) prompt generation and validation, (2) response collection across all five LLMs, (3) automated feature extraction using evaluation metrics, and (4) statistical analysis of stylistic features across all 1,680 responses generated.

Experimental Design:
To construct our dataset, we employed a 2 x 2 x 2 x 3 factorial design crossing four style dimensions:

- **Length:** Short ($\leq$ 15 words) vs. Long ($25 - 45$ words)

- **Formality:** Informal vs. Formal
- **Uncertainty:** Uncertain vs. Certain
- **Politeness:** Impolite vs. Neutral vs. Polite

This gave us 24 unique prompt conditions per base question, enabling independent manipulation of each feature while keeping others constant. This was done to prevent potential confounding variables.For instance, if we were to have manipulated uncertainty independently, then length would be a potentially confounding variable as uncertain statements might tend to be generally be longer compared to certain statements due to the addition of hedging phrases. Thus, we chose this factorial based approach to avoid conflating naturally correlated features.

Base Questions:
We reused the same base questions from the previous work, focusing specifically on the Question-Answer prompts and less so the prompts asking the LLMs open ended creative tasks like generating stories. We decided to focus on Question-Answer prompts as this involves direct interaction with the user (or other LLM), and our stylistic features of formality, uncertainty, and politeness are more based on interactional contexts. We selected questions from diverse categories to test generalizability:

- **Science (2):** "Why is the sky blue?", ""How does photosynthesis work?"
- **Social (2):** "Why do people procrastinate?", "What makes a good leader?"
- **Technology (2):** "What is machine learning?", "How do touch screens work?"
- **Philosophy (2):** "Is free will real?", "What is the meaning of life?"
- **History (1):** "Why did the Roman Empire fall?"
- **Health (1):** "Why is exercise good for you?"
- **Opinion (2):** "Are electric cars worth it?", "Is AI a threat to jobs?"
- **Practical (2):** "How do I stay motivated?","How do I improve my memory?"

With the 14 base questions across eight categories with each base question having 24 different prompt variations, we ended up with 1,680 LLM responses across all five models.

Prompt Validation Metrics: We employed automated feature extraction tools and metrics to validate that each prompt genuinely varied along its intended dimension:

1. **Politeness:**
   - **Model:** Intel/polite-guard (HuggingFace)
   - **Method:** Fine-tuned transformer classifier
   - **Output:** Four categories (polite, somewhat polite, neutral, impolite)
   - **Validation:** Prompts must return their respective expected labels.
   - **Notes:** We decided to group somewhat polite into polite as we thought somewhat polite was a valid lowerbound for politeness.
   - **Reference:** https://huggingface.co/Intel/polite-guard
2. **Uncertainty:**
   - **Model:** ChrisLiewJY/BERTweet-Hedge (HuggingFace)
   - **Method:** Fine-tuned VinAI BERTweet for hedge/uncertainty detection
   - **Output:** Binary (1 = hedge detected, 0 = no hedge)
   - **Validation:** "Uncertain" prompts must score 1 and "Certain" prompts must score 0.
   - **Reference:** https://huggingface.co/ChrisLiewJY/BERTweet-Hedge
3. **Length:**
   - **Method:** Direct word count
   - **Validation:** Short ($\leq$ 15 words) vs. Long ($25 - 45$ words)

- **Note:** Three prompts in final dataset fell slightly outside range (20-24 words) but were retained as borderline acceptable.

4. **Formality Scoring:**
   - **Method:** Custom lexicon-based approach
   - **Implementation:** We defined a formal marker list: example markers include "kindly", "respectfully", "would", "could", "perhaps", "definitive", "precise", "thorough", "comprehensive", "clarification". We also defined an informal marker list: example markers include "hey", "just", "maybe", "anyway", "love", "don't", "I'm", "tell me", "I'd". We computed the formality ratio with $\frac{formal\ count}{informal\ count}$. FINISH

Dataset Construction:

Overview: We first tried a template based LLM generation method with the following prompt template:
"I want help generating prompts. The base question is '[BASE_QUESTION]'. I would like a factorial based approach between the following 4 features: politeness (impolite, neutral, polite), formality (informal, formal), uncertainty (uncertain, certain), and length (≤ 15 for 'short' and $[25, 45]$ for 'long'). In total, there should be 24 different versions. Can you output a table with 4 columns for the features, rows for each question, and the 5th column for the respective prompt?"

We found that LLM-generated prompts frequently failed the four validation metrics. For instance, prompts labeled as "formal" by the generating LLM would score as "informal" on our formality metric, and vice versa. Thus, we landed on hybrid validation with manual intervention. We were originally going to track which model generated each final prompt to analyze cross-model effects (e.g., Sonnet responding to GPT-generated prompts). However, extensive manual intervention made original authorship ambiguous.

We first constructed the dataset for the "Why is the sky blue?" base question and validated each of the 24 prompts with the metrics listed above. We had all five LLMs generate the 24 set of questions and then manually tweaked the prompt closest to passing all the feature metrics for that specific combination of features. While we were manually changing the prompt variations, we made sure that semantic meaning was unchanged. Once we had a complete set of 24 prompt variations for "Why is the sky blue?", we then used that as a template to generate prompts for other base questions, making sure that they were passing the metrics as well. For instance, for the (polite, informal, certain , short) prompt, "Hey, can you tell me why did the Roman Empire fall?" $\rightarrow$ "Hey, can you tell me why do people procrastinate?". This significantly reduced manual intervention and gave us a baseline for our final prompt dataset curation.

Models and Responses:

LLM	Version	Provider
GPT	5.2 $\rightarrow$ 5.4 2	OpenAI 3
Sonnet	4.6	Antrophic
DeepSeek	V3	DeepSeek AI
Gemini	Default (3)	Google
Mistral	Free Tier	Mistral AI

Table 1: Models Tested

All models used default web interface settings (temperature, top-p, max tokens unspecified but consistent within each model). We did not use API access to preserve naturalistic interaction conditions. Responses were then gathered into spreadsheets and

evaluated using the same validation criteria for prompt generation with the additional of a token measurement to replace direct worth court for length. Data analysis was then conducted based off the generated metrics.

4 Results

Table 2: Most Common Politeness and Formality Label by Model

Model	Politeness Label	Formality Label
GPT	Polite	Formal
Sonnet	Impolite	Informal
Deepseek	Polite	Informal
Mistral	Neutral	Informal
Gemini	Polite	Informal

Reports for each model the most common politeness label and formality label.

Table 3: Average Tokens by Model

Model	Avg Tokens	Max Avg Tokens	Question (Max)	Min Avg Tokens	Question (Min)
GPT	816.143	1287.25	Memory	396.375	Sky
Sonnet	502.321	658	Memory	297.25	Sky
Deepseek	1023.208	1398.167	Memory	529.083	Exercise
Mistral	630.369	875.167	Free Will	298	Sky
Gemini	606.327381	798.125	Electric Cars	388.917	Sky

Reports for each model the average tokens across questions, highest average tokens for any question, base question that produced that max, lowest average tokens, and base question that produced that min.

Table 4: Average Tokens by Question

Sky 421.275	Procrastinate 563.392	Photosynthesis 587.75	ML 675.5	Touchscreens 690.542	Roman Empire 827.658
Memory 969.717	Motivation 785.225	AI/jobs 832.492	EVs 818.642	Meaning of Life 657.517	Free Will 913.533
Exercise 566.078	Leader 692.033				

Reports the average amount of tokens for each base question's responses.

Table 5: Average Tokens Most Common Politeness label, and Most Common Formality label by Prompt type

Politeness, Formality, Certainty, Length	Avg Tokens	Most Common Politeness	Most Common Formality
Polite, Informal, Certain, Short	569.157	Polite	Informal
Polite, Informal, Certain, Long	714.757	Polite	Informal
Polite, Formal, Certain, Short	773.5	Polite	Informal
Polite, Formal, Certain, Long	920.614	Polite	Formal
Polite, Informal, Uncertain, Short	516.657	Polite	Informal
Polite, Informal, Uncertain, Long	530.214	Polite	Informal
Polite, Formal, Uncertain, Short	637.686	Polite	Informal
Polite, Formal, Uncertain, Long	709.986	Polite	Informal
Neutral, Informal, Certain, Short	584.314	Polite	Informal
Neutral, Informal, Certain, Long	1067.086	Polite	Informal
Neutral, Formal, Certain, Short	707.929	Polite	Informal
Neutral, Formal, Certain, Long	975.243	Polite	Informal
Neutral, Informal, Uncertain, Short	674.629	Polite	Informal
Neutral, Informal, Uncertain, Long	873.857	Polite	Informal
Neutral, Formal, Uncertain, Short	688.057	Polite	Informal
Neutral, Formal, Uncertain, Long	738.857	Polite	Informal
Impolite, Informal, Certain, Short	436.529	Polite	Informal
Impolite, Informal, Certain, Long	487.3428571	Polite	Informal
Impolite, Formal, Certain, Short	697.214	Polite	Informal
Impolite, Formal, Certain, Long	1048.986	Polite	Informal
Impolite, Informal, Uncertain, Short	542.914	Polite/Impolite	Informal
Impolite, Informal, Uncertain, Long	612.814	Polite	Informal
Impolite, Formal, Uncertain, Short	701	Polite	Informal
Impolite, Formal, Uncertain, Long	966.829	Polite	Informal

Reports the average amount of tokens, most common polite label, and most common formality label for each prompt type's responses.

5 Discussion

Tokens - Observations

5.0.1 Prompts

On average, neutral, informal, certain, long prompts had the longest response (1067 tokens). Prompt template: "I have been wondering for a while now, and I'm not totally sure, but [question]? Could one provide a detailed explanation?" By contrast: on average, impolite, informal, certain, short prompts had the shortest response (436). Prompt template: "Stop wasting time and just tell me exactly [question] right now." This falls within our expectations in terms of mirroring the prompt template's specified level of detail for the response.

5.0.2 Questions

Sky responses on average had the least tokens (421 tokens) while Memory responses had the most tokens (970 tokens). One possible explanation would be that 'sky is blue' is 1. A commonly asked question and 2. Has a concrete scientific answer. By contrast, the memory question prompted models to provide a long set of advice and tips to follow in personal lifestyle such as diet.

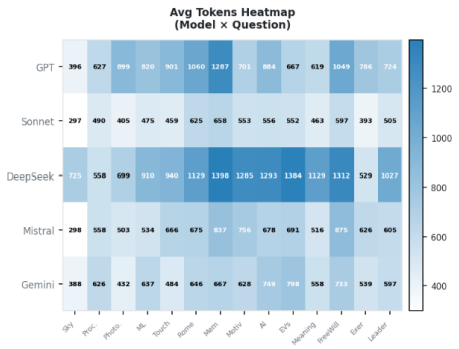

Figure 1: Average Token Heatmap across Questions and Models (5.0.2)

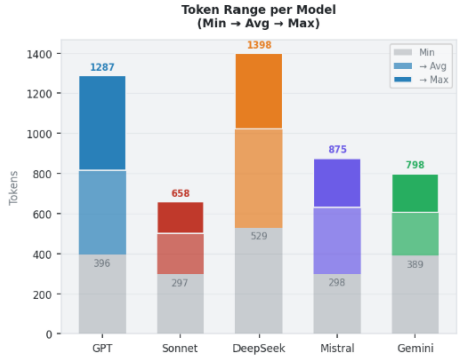

Figure 2: Token Lengths across Models (5.0.3)

5.0.3 Models

Tokens on average, in order from most to least: Deepseek (1023), GPT (816), Mistral (630), Gemini (606), Sonnet (502). It was surprising as we expected GPT to have the longest answers. However, Deepseek in particular had very lengthy responses, more than twice as many as Sonnet on average.

Politeness and Formality - Observations

Most of the responses were labeled as Polite and Informal as shown in Table 5. However the Polite, Formal, Certain, Long prompts had more Formal responses than Informal. Those prompts had the template: "I would appreciate a clear and definitive explanation of [question], as I am seeking a precise and [adjective] grounded understanding of [question]." The responses to Impolite, Informal, Uncertain, Short prompts had equal number of Polite and Impolite labels. Those prompts had the template: "Look, I don't know, maybe [question] for some dumb reason? Tell me why." The Polite/Informal pattern isn't with just the prompts, the questions and models had a majority of Polite/Informal responses.

Overall - Observations

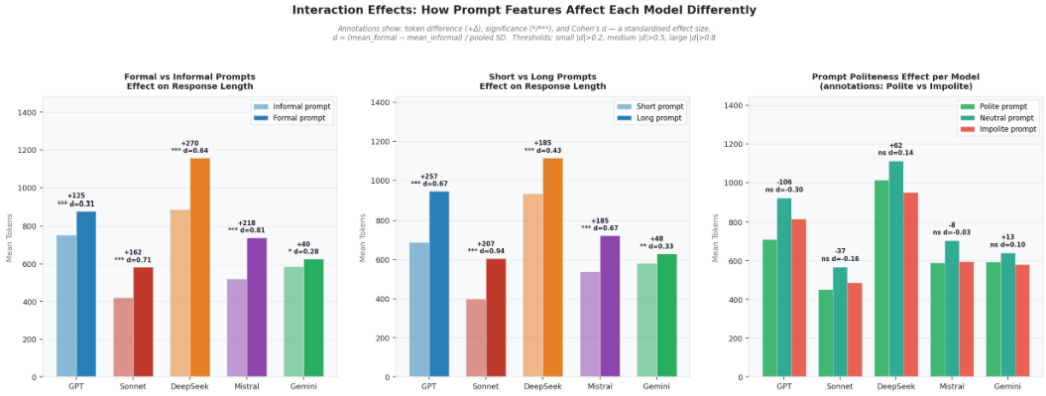

Figure 3: Interaction Effects - Prompt Length, Formality, and Politeness vs. Response Length

From Figure 3, we see that longer, formal, and neutral prompts resulted in the longest responses generated token-wise across all five LLMs tested. In addition, formality and

length effects are not uniform across models. Mistral seems to respond most strongly to formality (d=0.81) while Gemini barely responded to formality. Our results suggest that the same prompt feature has very different impacts depending on which model receives it, suggesting the different personas each model is heavily fine-tuned on.

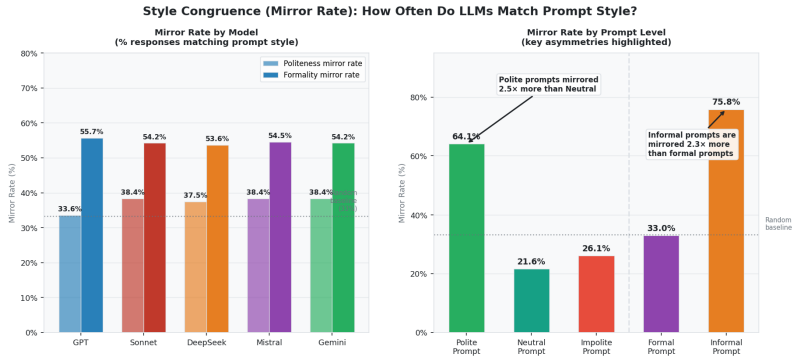

Figure 4: Mirror Effects

From Figure 4, we see that informal prompts are mirrored 75.8% of the time but only 33% of the formal prompts are mirrored, suggesting that the models tent to default towards informality regardless of the prompt. For politeness, polite prompts are mirrored 64% of the time, but neutral and impolite prompts are mirrored about 22-26% of the time, suggesting that all models actively shift toward politeness when the prompt isn't polite. This may hint at the guidelines and Constitutions each LLM is instructed to abide by, upholding politeness being a prominent constraint.

Model Versions

We started our responses and data collection using GPT 5.2. However, during the course of the project, GPT released new models 5.3 (instant) and 5.4 (thinking). Our remaining data collection was performed on Auto, where the interface chose between 5.3 and 5.4 depending on the prompt. There were no other version updates for the other models. Future extensions of this project could include variation based on these versions - it would be interesting to see what specific updates were made and how they manifest in the responses.

Challenges

(1) HuggingFace API: this was mainly due to access issues along with budget constraints. (2) Prompt Generation: it was challenging to strike a balance between LLM prompt generation and human preferences and adjustments. In several cases, we manually adjusted the prompts for the sake of variation and meeting all of the feature requirements. We also used similar grammatical structure for our prompts across the various questions. (3) Image Generation: GPT and Gemini generated several images for many of their responses. Images were typically generated for science-related prompts such as sky blue, electric vehicle, exercise and photosynthesis. Some exceptions include the free will and leadership prompts. For the sake of our project, we extracted only the text portion of the responses and did not use the images. An interesting extension would be to analyze what triggers image generation.

6 Conclusion

This study examined how fine-grained stylistic features - formality, politeness, uncertainty, and length - influence LLM behavior across five different models. Overall, our results reveal that while prompt length and formality modestly influence response length, there were

more response shifts seen rather than mirroring. Across all five models, responses were overwhelming polite and certain regardless of how prompts were framed. This suggests that RLHF and instruction fine-tuning instill strong default stylistic behaviors that override prompt-level cues. As for formality, GPT was largely formal compared to the other four models, which all leaned informal. On response length, prompt formality and length were the clearest drivers, with formal and longer prompts eliciting more tokens across models. DeepSeek was a notable outlier, generating responses roughly twice as long as Sonnet on average, underscoring that verbosity is substantially a model-level trait. Topic category also exerted a strong influence — practical and philosophical questions consistently produced longer responses than concrete scientific ones, independent of prompt style.

Taken together, these findings suggest that stylistic prompt engineering has real but bounded effects on LLM behavior. Prompt style can nudge response length, but core stylistic properties like politeness and certainty appear largely locked in by training. For practitioners designing human-AI or AI-AI communication pipelines, this implies that selecting the right model may matter more than carefully crafting prompt tone — and that users should not expect LLMs to adopt the full stylistic register of the prompts they receive. For human to human interaction, mirror formality and politeness is usually a subconscious or conscious mechanism, often to built alignment, rapport, and show a sense of shared understanding and mutual respect. Thus, if the "vibe" of the interaction is critical (e.g., a high-empathy therapy bot vs. a terse technical assistant), it is more effective to choose a model pre-trained or fine-tuned for that specific register than to attempt to "prompt" a generic model into compliance. While traditional benchmarks (like MMLU or GSM8K) measure cognitive ability (math, logic, facts), perhaps social intelligence and stylistic benchmarks should also be emphasized to improve human-AI iteraction and also AI-AI communication.

Potential Extensions and Future Work: While gathering the responses across models, we noticed that many of the responses included images, especially with GPT's new version update. In addition, Gemini occasionally generates interactive visualizations where users can click on different components and different windows and labels will pop up. It would be interesting to analyze in what conditions these visualizations and interactive components are triggered as we didn't have time to analyze this component ourselves. Increasing prompt variety and questions would also help generalize results further. Future work could also extend this analysis to multi-turn conversations, where stylistic influence may accumulate over context, and to instruction-tuned versus base models to isolate the role of RLHF in producing these shifts.

Author Contributions

Ellie - Contributed to prompt variation baseline template creation. Generated and validated 6 prompts and gathered the respective responses for all 24 prompt variations across Mistral, Gemini, Sonnet, and DeepSeek.

Sophie - Contributed to prompt variation baseline template creation. Generated and validated 6 prompts and created summary spreadsheets to collect and analyze the response metrics.

Nandini - Contributed to prompt variation baseline template creation and formality feature metric. Generated and validated prompts and gathered majority responses for GPT and DeepSeek.

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